# Unit-1

## [Introduction to Python and Jupyter Notebooks ]

| | |
|---|---|
| **Simple programs** | |
| 1 | Write a program to take two numbers as input and perform all basic arithmetic operations. |
| 2 | Write a program to calculate the area and perimeter of a rectangle using arithmetic operations. |
| 3 | Write a program to Calculate the Area of a Triangle |
| 4 | Write a program to Swap Two Variables. |
| 5 | Write a program to Convert Kilometers to meter, centimeters, inch, foot (1 km = 39370.1 inch, 1 km = 3280.84 feet) |
| 6 | Write a program to calculate the simple interest using the formula:<br>**SI = (P × R × T) / 100** |
| 7 | Write a program to convert temperature from Fahrenheit to Celsius using the formula:<br>**C = (F - 32) × 5/9** |
| 8 | Write a program to calculate the roots of a quadratic equation:<br>**$ax^2 + bx + c = 0$** |
| 9 | Write a program to convert a given number of seconds into hours, minutes, and remaining seconds. |
| 10 | Write a program to calculate the Body Mass Index (BMI) using the formula:<br>**BMI = weight / $height^2$** |
| **Programs on Conditional Operator, math library, Calender** | |
| 1 | Write a program to Check if a Number is Positive, Negative or 0. |
| 2 | Write a program to Check if a Number is Odd or Even. |
| 3 | Write a program to Find the Largest Among Three Numbers. |
| 4 | Write a program to Display Calendar. |
| 5 | Write a program to Find the Square Root |
| **If-else, if elif , Nested if** | |
| 1 | Write a program to check whether the given character is vowel or not. |
| 2 | Write a program to determine whether a number is positive, negative, or zero. |
| 3 | Write a program to assign grades based on marks:<br>• Marks >= 90: A<br>• Marks >= 80: B<br>• Marks >= 70: C<br>• Marks >= 60: D<br>• Marks < 60: F |
| 4 | Write a program to give weather advice based on temperature:<br>• Temperature > 35: "It's very hot!"<br>• Temperature between 20 and 35: "The weather is pleasant."<br>• Temperature < 20: "It's cold outside." |
| 5 | Write a program to check if a person is eligible for a driving license. |

<table>
<tr><td></td><td>

- The person must be at least 18 years old.
- They must pass both written and practical tests.

</td></tr>
<tr><td>6</td><td>Write a program to calculate electricity bills based on usage:<br><br>• First 100 units: $0.5/unit<br>• Next 100 units (101–200): $0.75/unit<br>• Beyond 200 units: $1.2/unit</td></tr>
<tr><td>7</td><td>Write a program to calculate the tax for an individual based on their salary.<br><br>• Salary less than $30,000: 10% tax<br>• Salary between $30,000 and $60,000: 15% tax<br>• Salary above $60,000: 20% tax</td></tr>
<tr><td>8</td><td>Write a menu driven program for arithmetic operations</td></tr>
<tr><td>9</td><td>Write a program to find maximum among three numbers.</td></tr>
<tr><td>10</td><td>Write a program to simulate a bank ATM. The user is prompted for their PIN. If the PIN is correct, display a menu to perform operations like checking balance, withdrawing money, or depositing money.</td></tr>
<tr><td>11</td><td>Calculate the late fee for returning a library book. Use the following rules:<br><br>• If returned within 5 days: $0.50 per day.<br>• If returned between 6-10 days: $1 per day.<br>• If returned after 10 days: $5 fixed penalty + $2 per additional day.</td></tr>
<tr><td>12</td><td>Check if a user is eligible for a job if age>=21, education is yes, and experience is greater than 2.prompt proper message if not eligible.</td></tr>
<tr><td>13</td><td>Write a program to simulate an order system where customers can choose a category (starter, main course, dessert), and then select an item. Each item has a price, and the program calculates the total bill.</td></tr>
<tr><td>14</td><td>Calculate shipping cost based on the weight of the package and the destination (domestic or international).<br><br>
<table>
<tr><td>Weight [Kg]</td><td>Domestic Price</td><td>International Price</td></tr>
<tr><td>< 1</td><td>5</td><td>15</td></tr>
<tr><td>2 to 5</td><td>10</td><td>25</td></tr>
<tr><td>6 to 10</td><td>25</td><td>50</td></tr>
<tr><td>>10</td><td>50</td><td>75</td></tr>
</table>
</td></tr>
<tr><td>15</td><td>Create a program that simulates a smart traffic light system. The program takes the following inputs:<br><br>1. Time of Day (morning, afternoon, evening, night).<br>2. Traffic Density (low, medium, high).<br>3. Emergency Vehicle Detected (yes or no).</td></tr>
</table>

| | |
|---|---|
| | Based on these inputs, decide how long the green light should stay on.<br><br>**Rules:**<br><br>1. If there's an emergency vehicle, the green light stays on for **60 seconds**, regardless of other inputs.<br>2. If it's **night** and traffic density is low, the green light stays on for **15 seconds**.<br>3. During other times:<br>  - Low traffic: **30 seconds**.<br>  - Medium traffic: **45 seconds**.<br>  - High traffic: **60 seconds**. |
| **Loop (range ,for while)** | |
| 1 | Write a program to display the multiplication Table. |
| 2 | Write a program to Find the Sum of all Natural Numbers till 100. |
| 3 | Write a program to calculate the sum of the digits of a number. |
| 4 | Write a program to reverse a number. |
| 5 | Write a program to check a number is palindrome or not. |
| 6 | Write a program to check a number is Armstrong number or not. |
| 7 | Write a program to create Countdown Timer using time module. |
| 8 | Write a program to check a number is prime number or not. |
| 9 | Write a program to print Fibonacci sequence |
| 10 | Write a program to find factorial of a number |
| 11 | WAP to take N numbers as input from the user and find the largest number using a loop. |
| 12 | Write a program to count a number of digits in a number. |
| 13 | Write a program to count Vowels in a String |
| 14 | Write a program to count Frequency of Characters in a String |
| 15 | Write a program with a loop that iterates over a list of numbers. For numbers less than 10, do nothing (use pass). |
| 16 | Write a program to print numbers from 1 to 10 but skip numbers that are divisible by 4. |
| 17 | Write a program that asks the user to enter numbers. If the user enters a negative number, exit the loop. |
| 18 | Write a program that prints all numbers from 1 to 20, skipping multiples of 3. |
| 19 | Create a program to play a Rock-Paper-Scissors game:<br><br>1. The user chooses Rock, Paper, or Scissors.<br>2. The computer makes a random choice.<br>3. The program determines the winner and displays the score after each round.<br>4. The game continues until the user decides to quit. |
| 20 | Create a Python program that allows a user to guess a randomly generated secret number between 1 and 100. The program should meet the following requirements:<br><br>1. The user has **5 attempts** to guess the number correctly. |

| | |
|---|---|
| | 2. After each guess, the program should provide feedback:<br>  - If the guess is **too low**, inform the user to guess higher.<br>  - If the guess is **too high**, inform the user to guess lower.<br>3. If the user guesses the number correctly within the attempts, display a success message and exit the loop.<br>4. If the user fails to guess the number after 5 attempts, display the secret number and a failure message.<br><br>**NOTE: Use random module in python** |